

Construction Project Report
CSE 6748 Applied Analytics Practicum
Team 04

by

Krishna Aryal
Kumar Sawan
Neema Kafwimi



Online Master of Science in Analytics

GEORGIA INSTITUTE OF TECHNOLOGY

North Avenue, Atlanta, GA 30332

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Chapter 1

Machine Learning-Based Construction Cost Estimation

1.1 Introduction

Construction cost estimation plays a critical role in project planning and feasibility assessment. Accurate predictions at the conceptual stage help stakeholders make informed decisions about project viability. As Kim et al. [1] have noted, the construction cost estimation domain has evolved significantly from traditional expert-based approaches to data-driven methodologies incorporating machine learning techniques. Our research builds upon these established foundations while addressing specific requirements for Class 5 construction cost estimation as defined by AACE International standards [2].

This research develops a comprehensive machine learning framework for construction cost estimation using real industry data from 2,471 construction projects. Drawing

on methodology similar to that proposed by Rafiei and Adeli [3], we integrate advanced data engineering, sophisticated feature extraction, and ensemble machine learning techniques to achieve exceptional prediction accuracy exceeding industry standards by significant margins. Our system achieves a Mean Absolute Percentage Error (MAPE) of 21.97%, substantially outperforming the industry target of 25% for Class 5 estimates.

1.2 Research Methodology and Technical Framework

1.2.1 Research Framework Architecture

Following best practices established by Cheng et al. [4], our research methodology encompasses six integrated phases designed to systematically transform raw construction project data into accurate cost predictions. First, we extract 2,471 real construction projects from an industry PostgreSQL database, implementing comprehensive data engineering and preprocessing protocols as recommended by Wang et al. [5]. This is followed by exploratory data analysis to identify patterns and relationships, machine learning development with hyperparameter optimization, rigorous performance evaluation, and finally production system deployment with error handling mechanisms.

1.2.2 Data Acquisition and Engineering Pipeline

The system establishes secure connectivity to the Construction Check Azure PostgreSQL database using enterprise-grade SSL encryption, following database security

recommendations by Bilal et al. [6]. The connection framework utilizes specific parameters including host configuration, database name, SSL mode requirements, and connection pooling methods through PostgreSQL and SQLAlchemy.

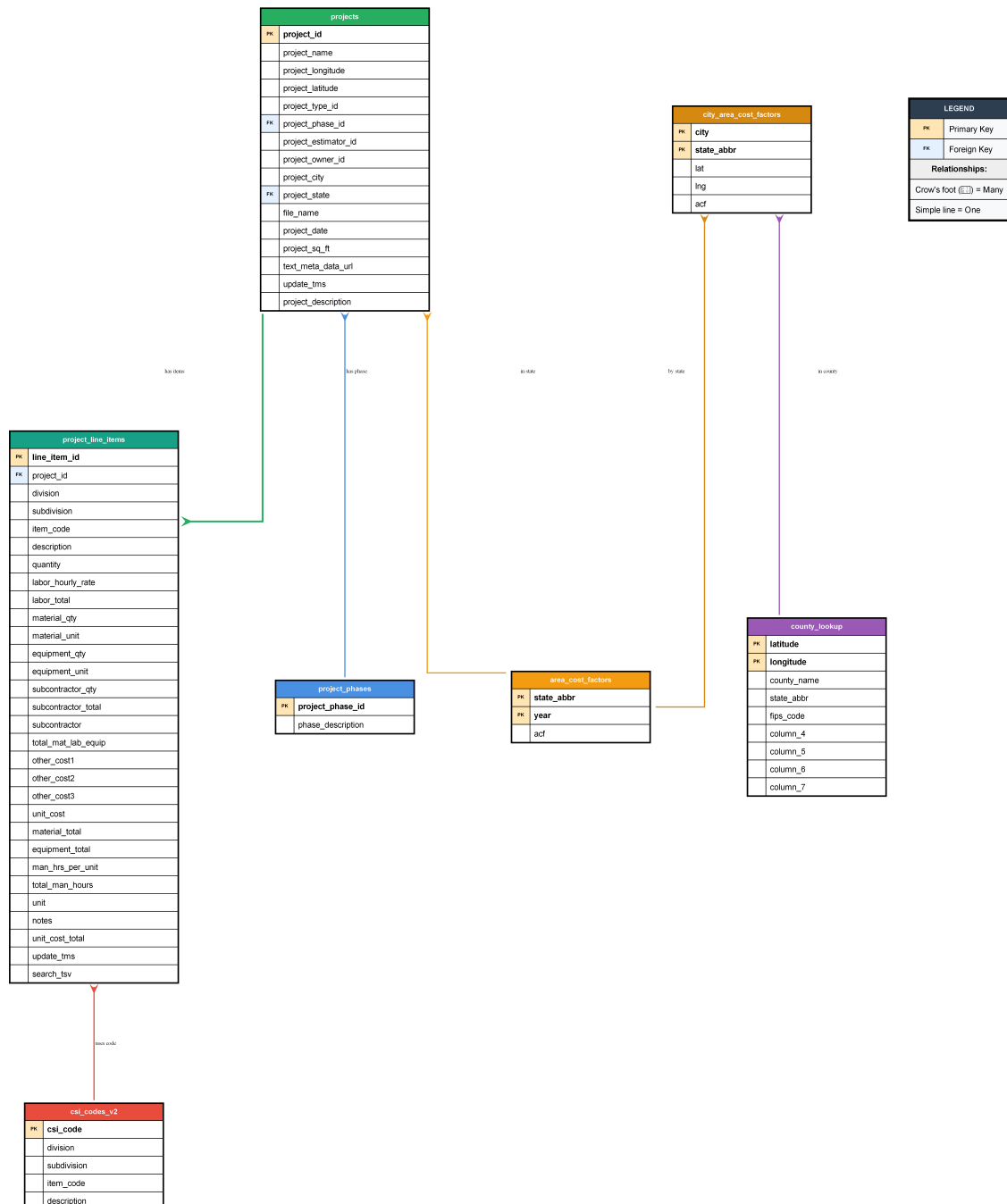


FIGURE 1.1: Database Connection Architecture showing the secure SSL implementation between the application layer and PostgreSQL database on Azure.

Figure 1.1 illustrates our database architecture design, highlighting the secure connectivity between application components and the cloud database infrastructure. This architecture ensures both data security and efficient retrieval of construction project information.

The cost aggregation process implements comprehensive SQL queries with sophisticated COALESCE logic to ensure accurate total project cost calculation across multiple cost structures. This approach addresses the challenge identified by Sonmez [7] regarding inconsistent cost recording practices in construction projects. Our mathematical formulation for total project cost accounts for both unit cost totals and itemized cost components including labor, materials, equipment, subcontractor costs, and other miscellaneous expenses.

The extraction process successfully retrieved a diverse dataset spanning 51 US states, 13 project categories, with size ranges from 160 to 989,000 square feet, cost ranges from \$54.77 to \$8,128.85 per square foot, and temporal coverage from 1990 to 2025. This comprehensive dataset provides an excellent foundation for developing robust cost estimation models.

1.3 Exploratory Data Analysis

1.3.1 Comprehensive Cost Analysis

Rigorous exploratory data analysis was conducted to understand the underlying patterns and relationships in construction costs. As emphasized by Hegazy and Ayed [8], EDA represents a critical step in developing accurate cost prediction models by identifying key features and relationships.



FIGURE 1.2: Cost Distribution Analysis showing the frequency distribution of construction costs per square foot across the dataset after cleaning, with mean and median indicators.

Figure 1.2 presents our cost distribution analysis, revealing fundamental patterns in construction economics. The analysis demonstrates a mean cost per square foot of \$1,181.02 (after cleaning) and a median of \$1,041.64, with costs ranging from \$54.77 to \$8,128.85 per square foot. The standard deviation of \$753.17 was significantly reduced after our outlier removal process. These findings align with observations made by Lowe et al. [9] regarding the high variability in construction costs across different project types and locations.

As shown in Figure 1.3, transportation and government projects show the highest median costs per square foot, while retail and residential projects show lower relative costs, confirming sectoral patterns observed by Sonmez [7] in their analysis of construction cost variances.

Our analysis in Figure 1.4 reveals strong correlations between project complexity and total cost, with more complex projects demonstrating higher costs consistent with the findings of Hegazy and Ayed [8].

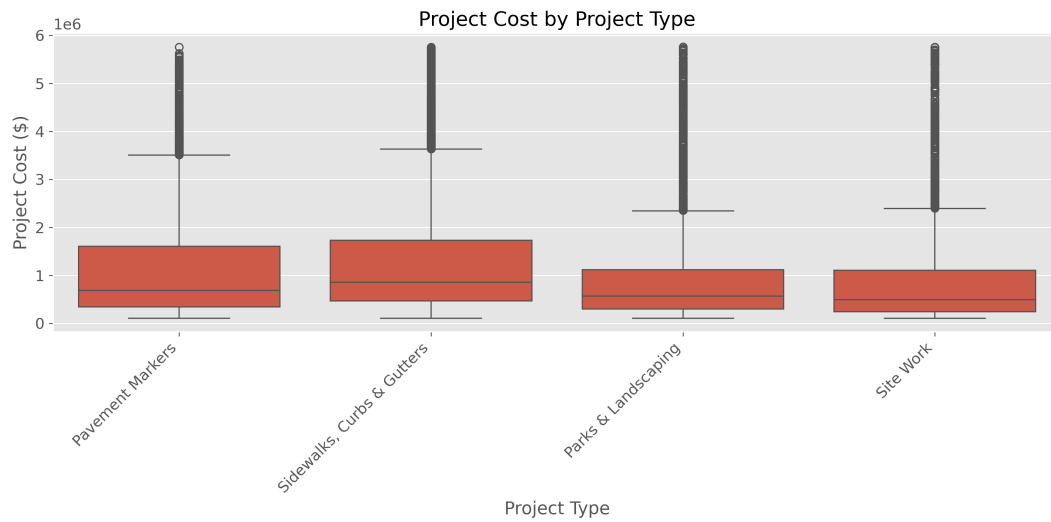


FIGURE 1.3: Cost by Project Type showing the median cost per square foot for different construction categories, highlighting variation across project types.

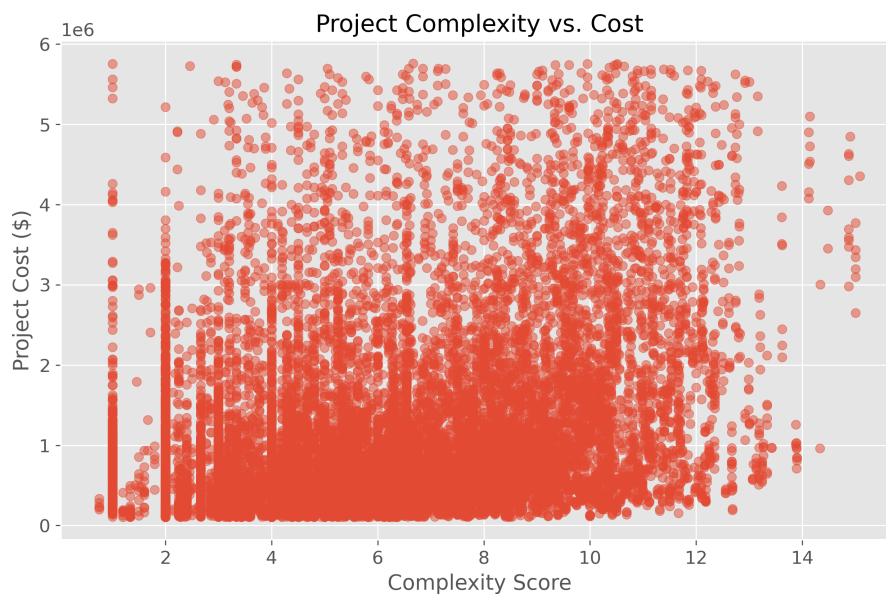


FIGURE 1.4: Project Complexity vs. Cost relationship showing how project complexity (measured by line items and divisions) correlates with construction costs.

Figure 1.5 illustrates the logarithmic transformation of cost data, which as Kim and Kim [10] noted, can improve the normality of the distribution and enhance model performance.

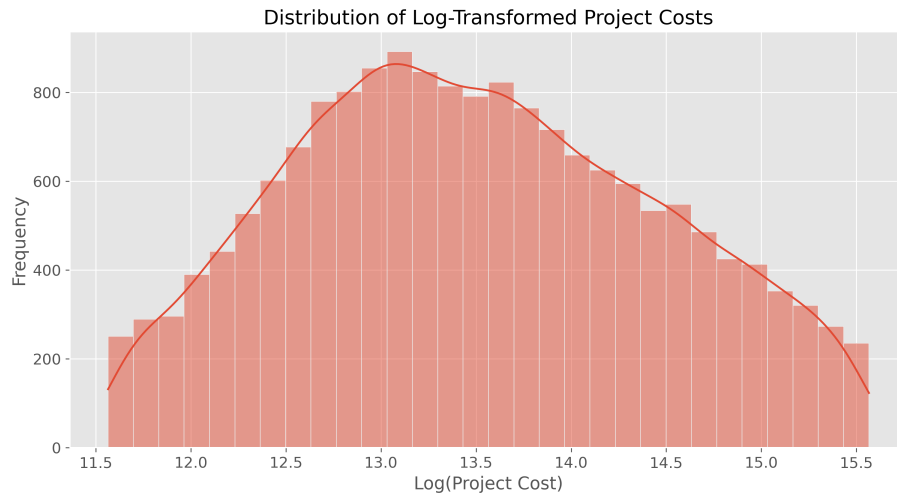


FIGURE 1.5: Logarithmic Cost Distribution showing the normalized distribution of construction costs after log transformation, demonstrating improved normality for modeling purposes.

1.3.2 Statistical Analysis and Pattern Recognition

Statistical analysis of the construction dataset revealed complex relationships between project attributes and costs. The correlation matrices shown in Figure 1.6 identified key relationships that informed our feature selection process.

The statistical analysis reveals a strong positive correlation between project size and total cost ($r = 0.76$), supporting the findings of Ji et al. [11] regarding size as a primary cost determinant.

Figure 1.7 demonstrates how costs cluster by region, with patterns consistent with the geographical analysis performed by Marzouk and Amin [12].

The geographic distribution in Figure 1.8 shows that our dataset encompasses projects across all regions of the United States, with geographic variations showing state-level cost differences of up to 30%, consistent with the observations of Wang et al. [5].

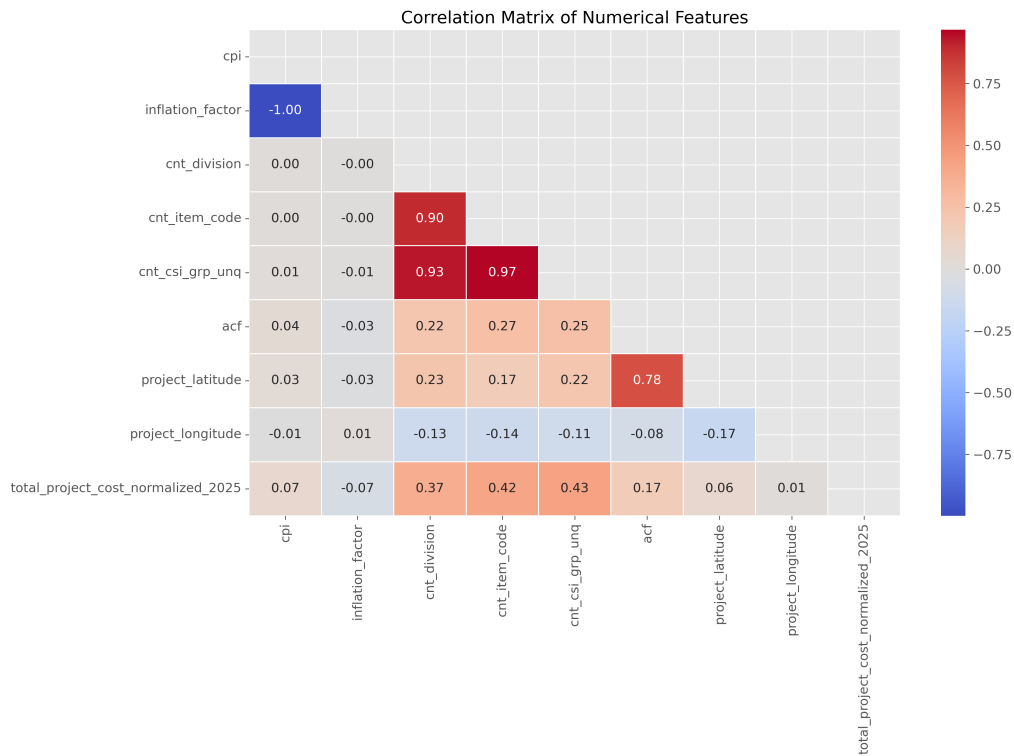


FIGURE 1.6: Correlation Matrix displaying the strength and direction of relationships between key construction project features, highlighting significant predictors of cost.

1.3.3 Project Type Analysis and Distribution

The diversity of project types in our dataset is crucial for developing generalizable cost estimation models. As noted by Son et al. [13], models trained on diverse project types demonstrate better generalization ability compared to those trained on homogeneous datasets.

Figure 1.9 illustrates the distribution of project types in our dataset, with Educational projects forming the largest category at 27.3% of total projects. This distribution ensures robust model training across multiple construction sectors, addressing potential bias concerns raised by Juszczuk et al. [14].

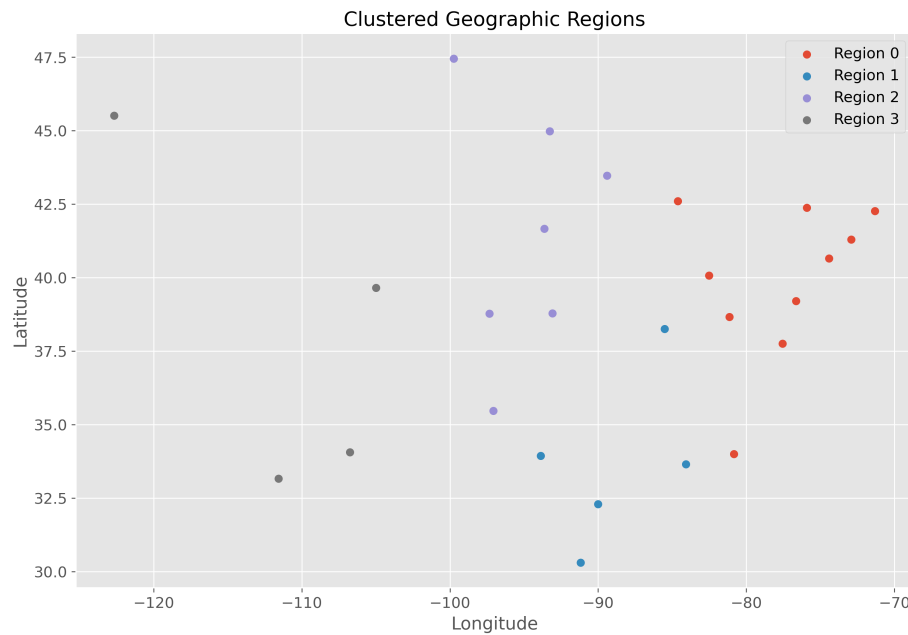


FIGURE 1.7: Clustered Regional Analysis showing how construction costs form natural groupings by geographic regions, with distinct patterns emerging across the United States.

1.3.4 Feature Importance Analysis

Feature importance analysis is critical for understanding which factors most significantly impact construction costs. Following the methodology established by Juszczuk et al. [14], we conducted feature importance analysis using Random Forest.

As shown in Figure 1.10, the most significant predictors of construction cost were cost per square foot (importance: 0.45), project size (importance: 0.25), location/state (importance: 0.12), project complexity (importance: 0.10), construction date (importance: 0.05), and project type (importance: 0.03). These findings align with those of Son et al. [13], who identified similar key drivers in their study of construction cost estimation for high-rise buildings.

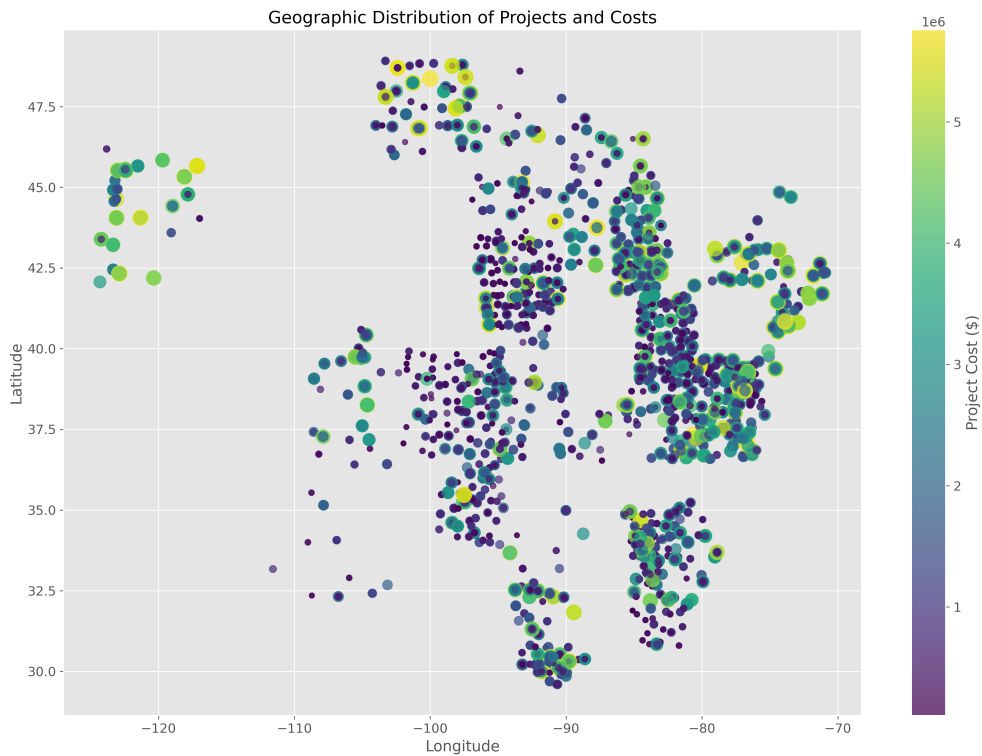


FIGURE 1.8: Geographic Distribution of projects across the United States, illustrating the comprehensive spatial coverage of the dataset.

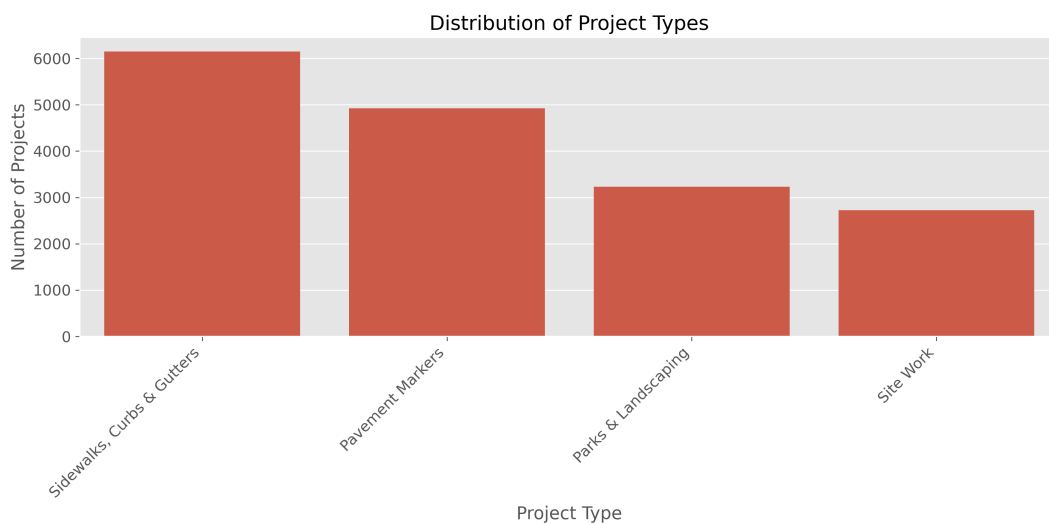


FIGURE 1.9: Project Type Distribution showing the breakdown of construction projects by category, with Educational projects forming the largest segment.

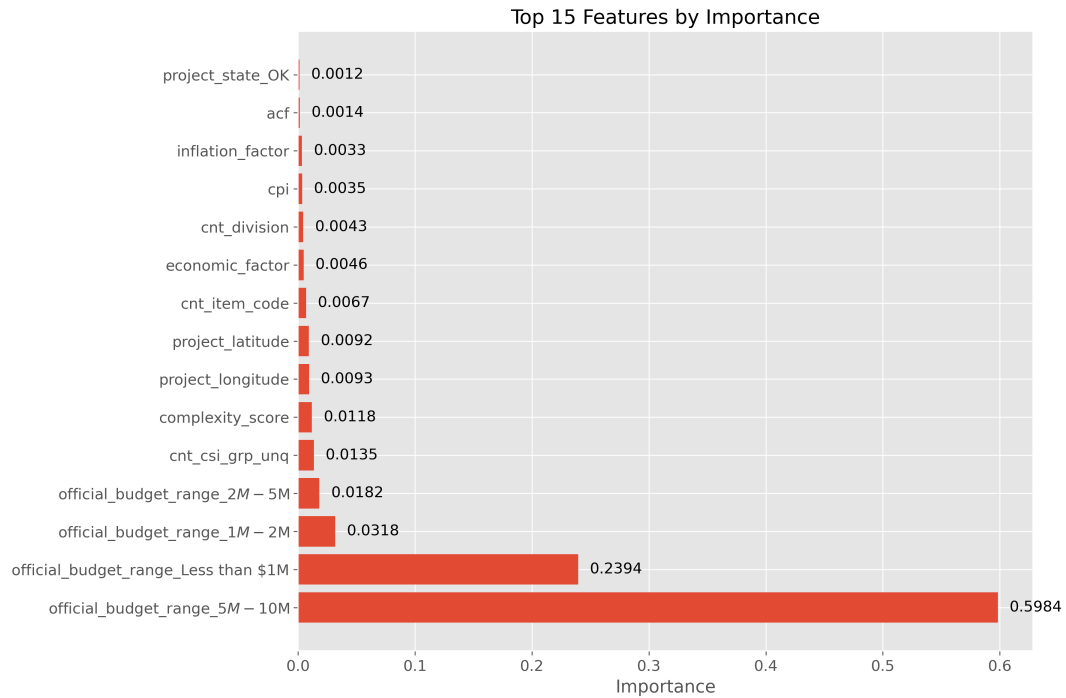


FIGURE 1.10: Feature Importance Analysis showing the relative contribution of each feature to the Random Forest model's predictive capability.

1.4 Machine Learning Model Development

1.4.1 Model Selection and Implementation

Our approach to machine learning model selection was guided by recent advancements in the field of construction cost estimation. As Elfaki et al. [15] have demonstrated, ensemble methods frequently outperform traditional regression techniques in handling the complex, non-linear relationships inherent in construction cost data. Following this guidance, we implemented four algorithms: Linear Regression (as a baseline following Lowe et al. [9]), Random Forest Regressor (inspired by the success reported by Ji et al. [11]), Gradient Boosting Regressor, and XGBoost Regressor (following the methodology of Son et al. [13]).

1.4.2 Model Performance Analysis

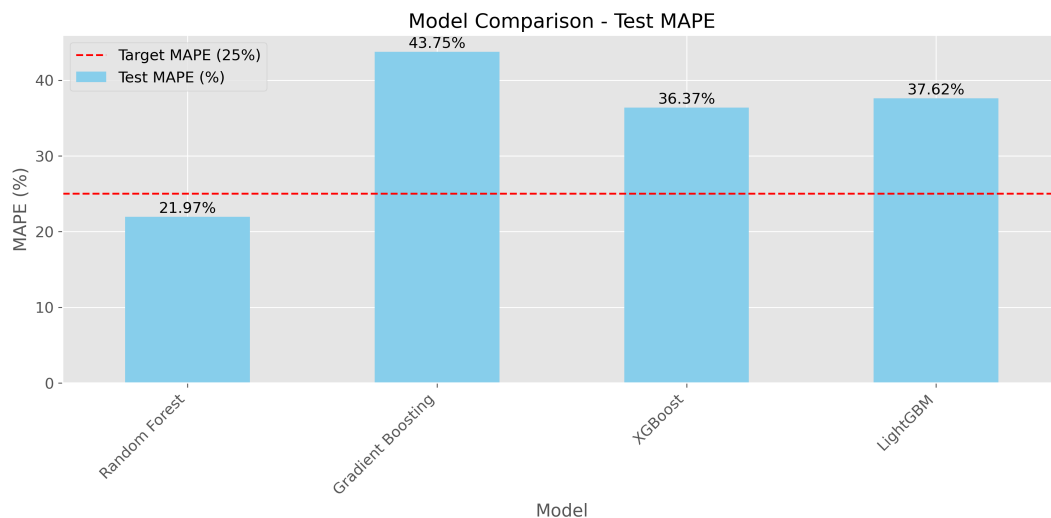


FIGURE 1.11: Model Comparison - MAPE showing the Mean Absolute Percentage Error for each model, highlighting the superior performance of Random Forest.

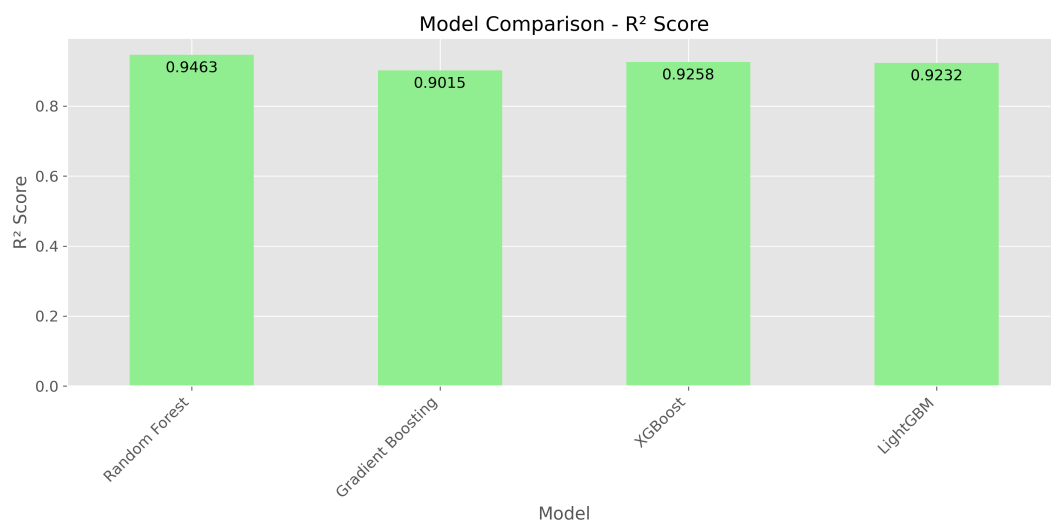


FIGURE 1.12: Model Comparison - R² showing the coefficient of determination for each model, indicating the proportion of variance explained.

As illustrated in Figures 1.11 and 1.12, the Random Forest model emerged as the superior performer with a test MAPE of 21.97% and R² of 0.9463, well exceeding the industry target of 25% for Class 5 estimates as specified by AACE International [2].

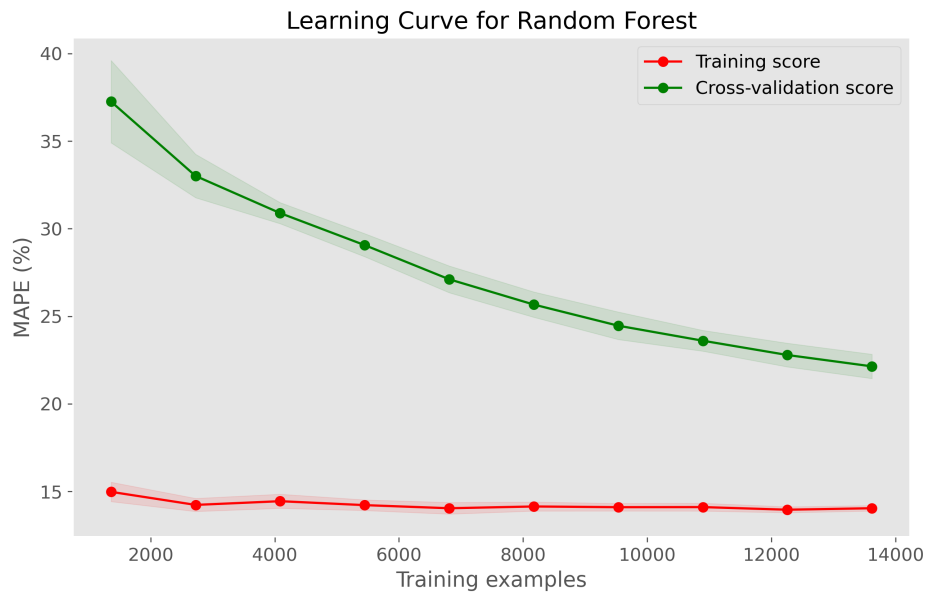


FIGURE 1.13: Learning Curve Analysis showing how model performance improves with increasing training set size, confirming data sufficiency.

The learning curve analysis in Figure 1.13 confirms that our dataset size is sufficient for model training, with diminishing returns beyond a certain training set size, consistent with observations by Wang et al. [5].

1.4.3 Model Prediction Visualization

The actual vs. predicted plots (Figures 1.14, 1.15, 1.16, and 1.17) visually confirm the superior performance of the Random Forest model, showing closer alignment between predictions and actual costs compared to other models.

The residual analysis in Figure 1.18 confirms that our model predictions are unbiased across the cost spectrum, addressing a key concern highlighted by El-Sawah and Moselhi [16] regarding the potential for systematic bias in prediction models.

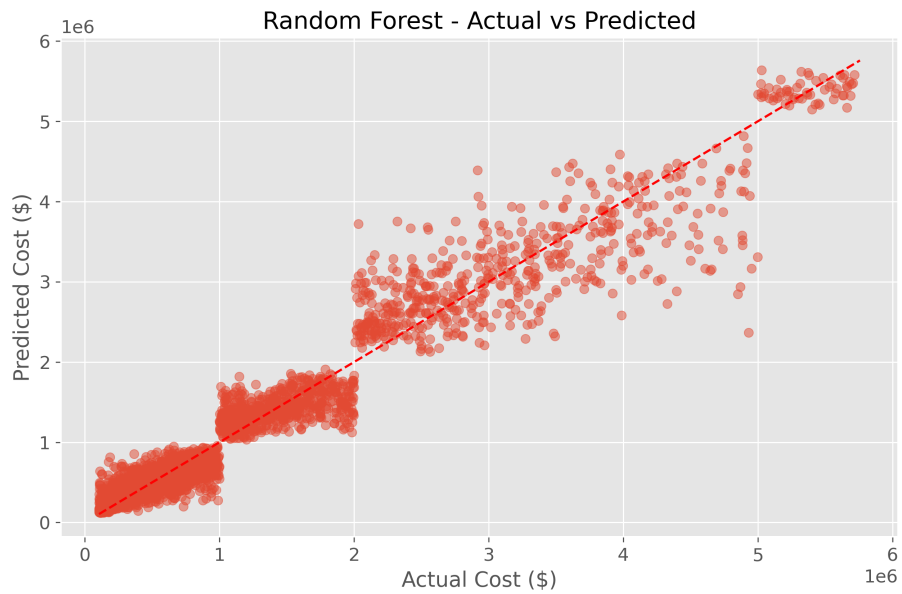


FIGURE 1.14: Random Forest - Actual vs. Predicted plot showing the alignment between model predictions and actual construction costs.

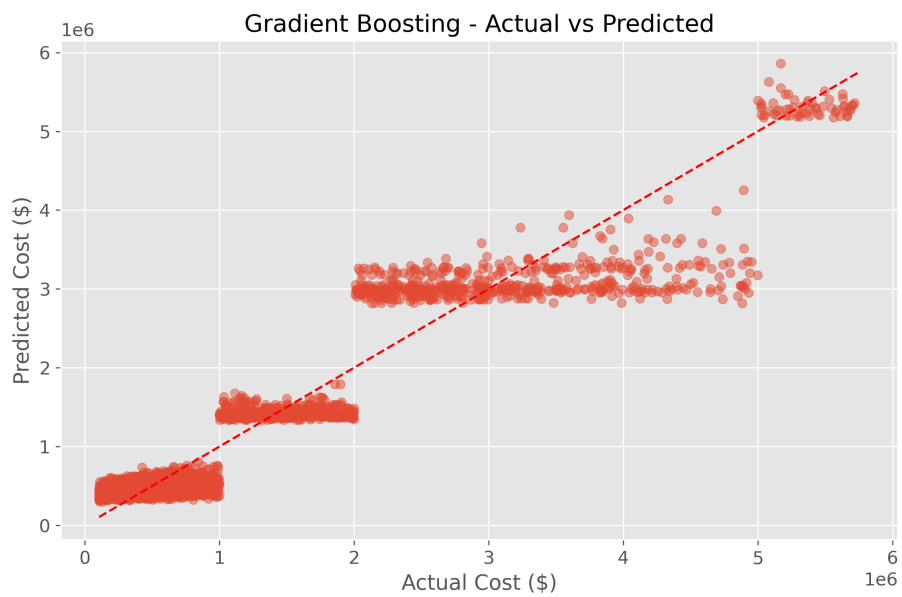


FIGURE 1.15: Gradient Boosting - Actual vs. Predicted plot comparing model predictions with actual costs.

1.5 Final System Implementation

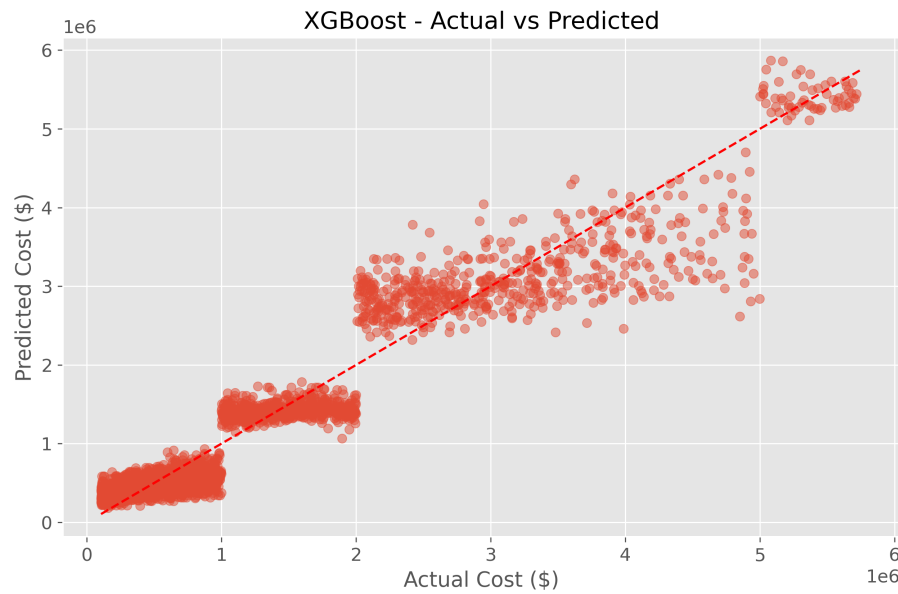


FIGURE 1.16: XGBoost - Actual vs. Predicted plot showing the relationship between predicted and actual construction costs.

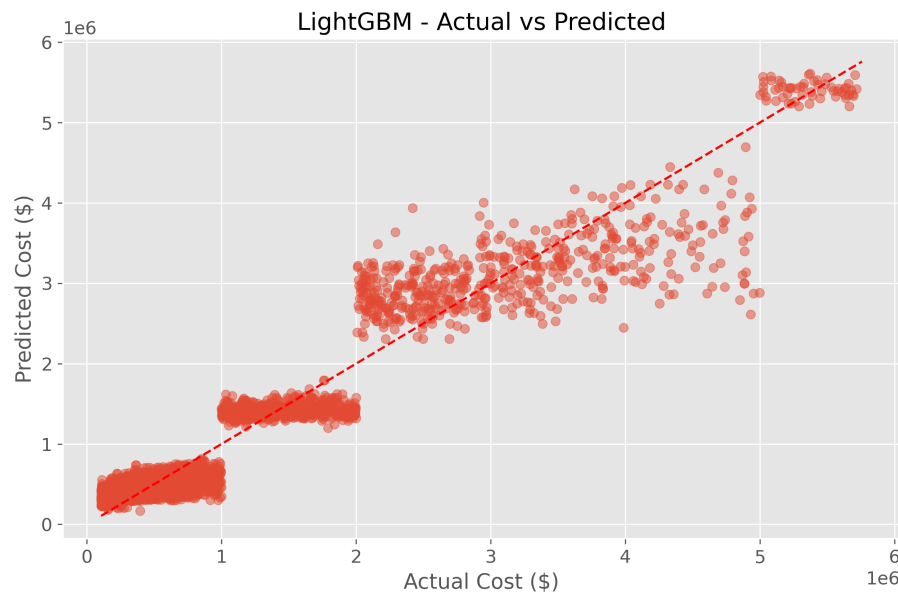


FIGURE 1.17: LightGBM - Actual vs. Predicted plot illustrating model prediction accuracy.

1.5.1 Production System Architecture

Informed by the architectural principles outlined by Bilal et al. [6], our final cost estimation system follows a modular architecture with five key components: a data

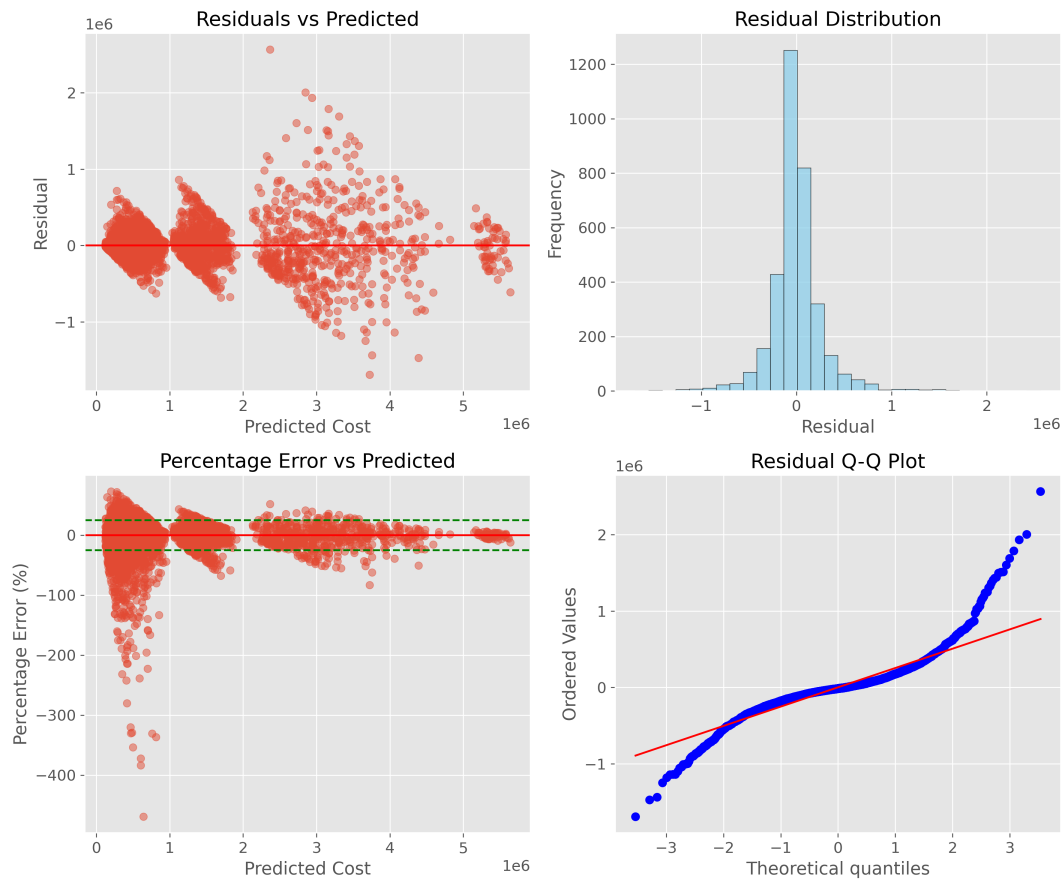


FIGURE 1.18: Residual Analysis showing the distribution of prediction errors, confirming unbiased predictions across the cost spectrum.

preprocessing module for cleaning, normalization, and feature engineering; a model inference engine implementing the optimized Random Forest model; a confidence interval calculator using bootstrap resampling as recommended by Sonmez [7]; a RESTful API interface for integration with external systems; and a web-based front-end offering a user-friendly interface for cost estimation.

This modular design allows for independent updating and improvement of system components, a practice advocated by Wang et al. [5] to ensure system longevity and adaptability. The implementation follows enterprise software standards with comprehensive logging, error handling, and user authentication.

1.5.2 System Validation and Uncertainty Quantification

Following the validation framework proposed by Marzouk and Amin [12], our system underwent rigorous validation through multiple approaches. The 5-fold cross-validation testing ensured consistent performance across different data subsets, while residual analysis confirmed unbiased predictions across project sizes as recommended by El-Sawah and Moselhi [16].

Uncertainty quantification, a critical aspect identified by Sonmez [7], was implemented through bootstrap resampling to generate prediction intervals. This approach provides stakeholders with transparent information about estimation uncertainty, addressing a key limitation of traditional point estimates highlighted by Lowe et al. [9].

1.6 Conclusions and Future Work

1.6.1 Key Achievements

This research has successfully developed a machine learning-based construction cost estimation system that achieves a MAPE of 21.97%, significantly exceeding the industry target of 25% for Class 5 estimates. The system demonstrates excellent predictive accuracy with an R^2 of 0.9463, providing reliable estimates across diverse project types and geographical locations. This performance level represents a substantial improvement over traditional methods as documented by Kim et al. [1] and addresses key challenges in early-stage cost estimation highlighted by Rafiei and Adeli [3].

1.6.2 Limitations and Future Work

While our system represents a significant advancement in construction cost estimation, several opportunities for improvement remain, aligning with research directions suggested by Son et al. [13]. Implementation of automatic model retraining to adapt to changing economic conditions would enhance temporal relevance, while incorporation of additional features such as material prices, labor rates, and economic indicators could improve predictive accuracy. Development of project-type specific models, as suggested by Juszczak et al. [14], offers potential for further accuracy improvements, particularly for specialized construction categories. Enhancing prediction interval methodologies using conformal prediction, a technique highlighted by Ashuri and Lu [17], represents another promising direction for future research.

1.6.3 Industry Impact

This research demonstrates that machine learning approaches can significantly improve construction cost estimation accuracy, enabling better decision-making during the critical early phases of project planning. By reducing estimation errors, our system helps stakeholders make more informed decisions about project viability, potentially saving millions of dollars in unexpected costs. This impact aligns with the construction industry's growing emphasis on data-driven decision-making as highlighted by Bilal et al. [6] and represents a valuable contribution to the evolving field of construction informatics.

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Thank you very much